

## Solution Requirements

|                      |                          |
|----------------------|--------------------------|
| <b>Date</b>          | 29 September 2026        |
| <b>Team ID</b>       | 6ab4cf016603573ccfb54f03 |
| <b>Project Name</b>  | AI-WeatherWise           |
| <b>Maximum Marks</b> | 4 Marks                  |

### Define Functional and Non-Functional Requirements:

Create a comprehensive list of requirements to define exactly what your solution will do and how well it will perform.

- **Functional Requirements (FR)** define specific behaviors, functions, or features of the system (What the system should *do*).
- **Non-Functional Requirements (NFR)** specify the criteria that judge the operation of a system, rather than specific behaviors (How the system should *be*).

Fill out the descriptions and necessary details for each category provided below.

### Step 1: Functional Requirements (FR)

| S.No | Requirement Category | Requirement Description                                                                                                                                                      | Priority (High/Medium/Low) |
|------|----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------|
| 1    | Authentication       | The system should allow registered users to securely log in and access the weather application.                                                                              | High                       |
| 2    | Authorization levels | The system should provide appropriate access permissions for users and administrators based on their roles.                                                                  | Medium                     |
| 3    | External interfaces  | The system should integrate with a third-party <b>Weather API</b> to retrieve real-time weather information such as temperature, humidity, wind speed, and weather condition | High                       |

|   |                                   |                                                                                                                                                                      |        |
|---|-----------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|
| 4 | Transactions processing           | The system should process user weather requests, send them to the weather API, retrieve the required data, and return the processed weather information to the user. | High   |
| 5 | Reporting                         | The system should provide weather information and AI-generated insights in a clear and understandable format for the requested location.                             | Medium |
| 6 | Business rules, etc.              | The system should validate the location input, process valid weather data, and generate relevant AI-based recommendations based on the available weather conditions. | High   |
| 7 | Compliance to laws or regulations | The system should protect user data and follow applicable data privacy and security requirements while handling user information and API data.                       | Medium |
| 8 | <i>Other</i>                      | The system should provide proper error handling, API response validation, and database connectivity to ensure smooth operation of the application.                   | Medium |

## Step 2: Non-Functional Requirements (NFR)

| S.No | NFR Category | Requirement Description | Target Metric / Acceptance Criteria |
|------|--------------|-------------------------|-------------------------------------|
|      |              |                         |                                     |

|          |                            |                                                                                                                                                          |                                                                                                                                                                                 |
|----------|----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>1</b> | Performance & Speed        | The system should retrieve and process weather information within a reasonable response time.                                                            | The system should return weather information within <b>3 seconds</b> under normal network conditions and should respond to API requests efficiently without unnecessary delays. |
| <b>2</b> | Scalability                | The system should be capable of handling an increasing number of users and weather API requests without significant performance degradation.             | System should support at least <b>100 concurrent users</b> without major performance degradation.                                                                               |
| <b>3</b> | Security & Data Privacy    | The system should protect user data and API credentials using secure authentication, authorization, and appropriate data protection measures.            | API keys and sensitive information must be stored securely in <b>environment variables</b> and not exposed in source code.                                                      |
| <b>4</b> | Reliability & Availability | The system should provide consistent weather information and continue to function reliably while handling API, database, or network failures gracefully. | System should successfully handle <b>95%+ of valid weather requests</b> and provide proper error messages when the API is unavailable.                                          |
| <b>5</b> | Usability & Accessibility  | The system should provide a simple, user-friendly, and accessible interface that allows users to easily search and understand weather information.       | Weather information should be displayed clearly, and users should be able to obtain weather results within <b>3 seconds</b> under normal network conditions.                    |
| <b>6</b> | <i>Other</i>               | The system should be maintainable, well-structured, and compatible with commonly used browsers and devices.                                              |                                                                                                                                                                                 |